Energy Release and Dissipation in Nuclear Fission

Step-by-Step Notes on Section 1.5 of E. E. Lewis,
Fundamentals of Nuclear Reactor Physics

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October 23, 2026

Overview

A single fission of a uranium-235 nucleus releases about 200 MeV. This energy is shared among the fission products and is eventually turned into heat in the fuel. How fast, and how far from the fission site, that happens depends on whether the carrier is electrically charged or neutral. The example reaction used throughout is

$$n + {}^{235}_{92}\mathrm{U} \longrightarrow {}^{236}_{92}\mathrm{U}^{*} \longrightarrow {}^{141}_{56}\mathrm{Ba} + {}^{92}_{36}\mathrm{Kr} + 3n + \sim 200\,\mathrm{MeV}. \tag{1}$$

Step by Step

1. **A neutron hits U-235 in the fuel and the nucleus absorbs it, becoming excited U-236.**
   The neutron adds energy to the nucleus, so U-236 starts out "excited" (with extra energy) and unstable. Here "excited" refers to the *nucleus*, not the atom's electrons. When the neutron is absorbed, its binding energy (about 6.5 MeV) is released inside U-236, so the protons and neutrons sit in higher-energy states and the nucleus vibrates and deforms. This is more than U-236's fission barrier (about 6 MeV), so the nucleus stretches and splits. The electron shells are not involved: electron excitations are only eV-scale, while this energy is MeV-scale and stays entirely in the nucleus.

   Why does the nucleus absorb the neutron?

   - **No Coulomb barrier.** The neutron has no charge, so the protons do not repel it. It can reach the nucleus even with almost zero kinetic energy, which a proton could not do.
   - **The strong force pulls it in.** Within about 1 fm of the nucleus, the neutron feels the attractive nuclear potential, roughly 40–50 MeV deep.
   - **Slow neutrons are captured more easily.** A thermal neutron spends more time near the nucleus, so the capture cross section is large (roughly $\propto 1/v$; about 680 b in total for U-235 at thermal energy, of which about 585 b leads to fission). This is why reactors use moderators. Absorption is not guaranteed: the neutron can also just scatter.

   Force picture vs. energy picture. These are the same physics. An attractive force means the potential energy drops as the neutron moves in ($F = -dV/dr$), so being bound inside is lower in energy than being free. The net energy gained is $S_n \approx 6.5$ MeV, which is *not* the well depth (40–50 MeV): the lower levels are already full, so by the Pauli principle the new neutron lands

near the top, at the Fermi level, about 6.5 MeV below zero. This S_n does not disappear; it becomes the excitation energy of U-236.

2. The U-236 splits into Ba-141 + Kr-92, plus 2–3 prompt neutrons and prompt gammas.
"Prompt" means these come out instantly, within about 10^{-14} s, at the moment of the split. The opposite is "delayed": particles emitted later (seconds to years) when the fission products radioactively decay.

Why does the excited nucleus split?

- **Strong force:** pulls all nucleons together, but only over very short distances.
- **Coulomb force:** pushes the protons apart, and reaches over long distances.

Before the neutron arrives, the strong force wins, so U-235 holds together.

(i) The absorbed neutron's energy (~ 6.5 MeV) goes into the motion of the nucleons. They move faster.

(ii) Faster nucleons can move farther apart from each other.

(iii) When nucleons get farther apart, the strong force between them drops sharply, because it is short-range.

(iv) The Coulomb force between the protons drops only a little, because it is long-range.

(v) Once enough nucleons are far enough apart, Coulomb beats the strong force and pushes them into two groups. That is the split.

The Coulomb force does not increase after absorption: the neutron has no charge, so there are still 92 protons. What changes is that the strong force loses much more than Coulomb does.

3. The ~ 200 MeV is the total energy of one fission, not a separate product.
It is the energy budget (the Q-value) that the products share among them.

Where does it come from? Energy is conserved:

$$\underbrace{M_{\text{U-235}}c^2 + m_n c^2}_{\text{before}} = \underbrace{M_{\text{Ba}}c^2 + M_{\text{Kr}}c^2 + 3m_n c^2}_{\text{rest masses after}} + \underbrace{T_{\text{Ba}} + T_{\text{Kr}} + T_{3n} + E_\gamma}_{\text{energy released}} \tag{2}$$

The slow neutron's kinetic energy is negligible. The rest-mass energy before is about 219,900 MeV and after is about 219,700 MeV. The difference, about 0.1%, is Q. Counting the later beta decays, Ba and Kr end as Pr-141 and Zr-92:

$$Q = \left[M_{\text{U-235}} + m_n - M_{\text{Pr-141}} - M_{\text{Zr-92}} - 3m_n\right] c^2 \approx 200\,\text{MeV} \tag{3}$$

What Q includes: only energy beyond rest mass.

$$Q = \underbrace{T_{\text{Ba}} + T_{\text{Kr}}}_{168} + \underbrace{T_{3n}}_{5} + \underbrace{E_\gamma}_{7} + \underbrace{E_\beta + E_{\gamma,\text{delayed}} + E_{\bar{\nu}}}_{\text{8+7+12 (later, from beta decay)}} \approx 200\,\text{MeV} \tag{4}$$

All values are in MeV. Most of it, about 168 MeV, is simply Ba and Kr flying apart very fast.

4. Ba and Kr move so fast that they lose their outer electrons and become positive ions, each with about $+20e$ of charge.

In short: the fragments leave the split carrying the U atom's electrons, but they move too fast to keep the outer ones. Each fragment is left with about 20 fewer electrons than protons, so it is a positive ion.

Why the outer electrons are lost. The fragments move at about 10^7 m/s. An electron stays attached only if its orbital speed is greater than the speed of the ion, so the outer electrons cannot keep up and the fragment becomes a highly charged ion. A Bohr-model estimate of the orbital speed is $v \approx (Z_{\text{eff}}/n)\,\alpha c$, with $\alpha c \approx 2.2 \times 10^6$ m/s:

	Ba ($Z = 56$)	Kr ($Z = 36$)
Innermost (1s) electron	$\sim 1.2 \times 10^8$ m/s ($\sim 0.4c$)	$\sim 7.9 \times 10^7$ m/s ($\sim 0.26c$)
Middle shells ($n = 4$ Ba, $n = 3$ Kr)	$\sim 1 \times 10^7$ m/s	$\sim 1.5 \times 10^7$ m/s
Outermost (valence) electrons	$\sim 2 \times 10^6$ m/s	$\sim 2 \times 10^6$ m/s
Fragment speed	$\boldsymbol{\sim 1.0 \times 10^7}$ **m/s**	$\boldsymbol{\sim 1.4 \times 10^7}$ **m/s**

The inner electrons orbit 5–10 times faster than the fragment moves, so they stay attached. The outer electrons are 5–7 times slower, so they are stripped. The cutoff falls in the middle shells, which is why about 20 electrons are lost from each fragment. The fragment speeds follow from the kinetic energy split (about 100 MeV for Kr and 68 MeV for Ba): the lighter Kr is faster, so it ends up slightly more stripped. The original U atom had 92 electrons. The fragments keep their tightly bound inner electrons, and the stripped outer ones (roughly 40 in total) are left behind at the fission site (what happens to them is covered in step 6).

5. Because they are positively charged, the fast Ba and Kr ions strip electrons off the U and O atoms of the fuel, losing energy each time.

The fuel is UO_2, a solid crystal of uranium and oxygen atoms.

(i) The Ba and Kr ions fly through this crystal.

(ii) Their positive charge ($+20e$) pulls on the electrons of the U and O atoms they pass (Coulomb force) and knocks those electrons free. The U and O atoms become ionized.

(iii) The fragments do *not* keep these electrons; they knock them loose and keep moving.

(iv) Each electron freed costs the fragment a little kinetic energy (about 30 eV), so the fragment slows down.

(v) Millions of these small losses stop the fragment within a few microns.

The freed electrons and ionized atoms are what turn the energy into heat (step 6).[1]

6. The ionized U and O atoms get their electrons back, and the energy becomes heat, right at the fission site.

"Ionized atoms" are the U and O atoms from step 5, and "free electrons" are the electrons knocked out of them, not electrons from the fission. Each fragment frees roughly 3 million

[1] Only near the very end of its path, when it is slow, does the fragment get close enough to the U and O nuclei for its own nucleus to push directly on theirs. These collisions can knock U or O atoms out of their places in the crystal, which is a form of radiation damage to the fuel. They account for only a small fraction of the fragment's energy loss.

electrons along its path ($\sim 80\,\mathrm{MeV} \div \sim 30\,\mathrm{eV}$ per electron). Almost all of them go back to the ionized U and O atoms.

Why does a round trip make heat? The electron's trip out and back is not the point; the electron is only a middleman that carries energy from the fragment to the crystal. Follow the energy:

- The fragment does work to pull the electron off, so the fragment slows down. That energy leaves the fragment for good.
- The freed electron now holds that energy: partly as the energy it took to free it, partly as its own motion.
- The electron bumps into nearby atoms as it slows, then drops back into a hole. Each step hands its energy to the atoms around it, making them vibrate.
- That vibration of the crystal is heat.

The electron ends where it started, so for the electron the round trip nets to zero. But the fragment is now slower and the crystal is hotter: the net effect is fragment motion turned into crystal vibration. (Rubbing your hands is similar: nothing leaves your hands, yet the motion of your arms becomes warmth.)

The ~ 40 electrons left behind at the fission site (step 4) do the same: they slow down and recombine with these ionized atoms. UO_2 has no spare room for extra electrons; "absorbed" means the free electrons fill the holes the fragments created along their tracks. This is why the fuel itself is where the heat is produced.

7. Meanwhile, within a few microns the fragment stops, picking up about 20 electrons until it is a neutral atom.

Of the ~ 3 million freed electrons, only about 20 end up on the fragment; this is a small side effect alongside steps 5–6. The fragment captures electrons because it has a large positive charge ($+20e$) and empty electron orbits, so it attracts electrons much more strongly than a neutral atom holds its outer ones. Capturing an electron puts it in a lower-energy, more tightly bound state, which is favorable. While the fragment is very fast, it cannot hold onto them (step 4), so most of the capture happens near the end, when it is slow. Overall charge balances out: the ~ 40 electrons left behind at the fission site (step 4) join the UO_2, and the fragments take about the same number from the UO_2 along their paths.

8. The fragments are radioactive and beta-decay to stable nuclei.

Why the fragments are radioactive: they have too many neutrons. U-236 has 144 neutrons for 92 protons, a ratio of about 1.57; heavy nuclei need that many neutrons to stay together. When it splits, the fragments inherit roughly the same ratio. Lighter nuclei, however, are stable with fewer neutrons per proton (about 1.3–1.4). For example:

- **Ba-141** has 56 protons and 85 neutrons, but the heaviest stable barium is Ba-138 (82 neutrons).
- **Kr-92** has 36 protons and 56 neutrons, but the heaviest stable krypton is Kr-86 (50 neutrons).

Both fragments therefore have extra neutrons. Beta decay fixes this by turning a neutron into a proton, one step at a time:

$$n \longrightarrow p + e^{-} + \bar{\nu}_e. \tag{5}$$

When does the beta decay happen? Almost always after the fragment has stopped and become a neutral atom in the fuel. It does not wait for this; the decay is random and its timing is set only by the half-life. The decay is a nuclear process, so it does not depend on whether the atom is moving or charged. Stopping is simply much faster:

- Stopping time: a few microns at $\sim 10^7\,\mathrm{m/s}$, about $10^{-12}\,\mathrm{s}$.
- Half-lives: Kr-92 $\sim 1.8\,\mathrm{s}$, Ba-141 $\sim 18\,\mathrm{min}$, and later steps in the chains take hours to about a month (Ce-141 $\sim 33\,\mathrm{days}$).

The chance that Kr-92 decays during that picosecond is about $10^{-12}/1.8 \approx 4\times10^{-13}$, essentially zero.

How they decay: each fragment goes through a beta-decay chain to a stable nucleus, emitting betas, gammas and antineutrinos. In each step, A (the mass number) stays the same and Z (the number of protons) goes up by 1. The new nucleus is often left excited and emits a gamma (γ).

Barium chain:

$$^{141}_{56}\mathrm{Ba} \longrightarrow {}^{141}_{57}\mathrm{La} + e^- + \bar{\nu}_e\ (+\gamma) \tag{6}$$
$$^{141}_{57}\mathrm{La} \longrightarrow {}^{141}_{58}\mathrm{Ce} + e^- + \bar{\nu}_e\ (+\gamma) \tag{7}$$
$$^{141}_{58}\mathrm{Ce} \longrightarrow {}^{141}_{59}\mathrm{Pr} + e^- + \bar{\nu}_e\ (+\gamma) \qquad \text{(Pr-141 is stable)} \tag{8}$$

Krypton chain:

$$^{92}_{36}\mathrm{Kr} \longrightarrow {}^{92}_{37}\mathrm{Rb} + e^- + \bar{\nu}_e\ (+\gamma) \tag{9}$$
$$^{92}_{37}\mathrm{Rb} \longrightarrow {}^{92}_{38}\mathrm{Sr} + e^- + \bar{\nu}_e\ (+\gamma) \tag{10}$$
$$^{92}_{38}\mathrm{Sr} \longrightarrow {}^{92}_{39}\mathrm{Y} + e^- + \bar{\nu}_e\ (+\gamma) \tag{11}$$
$$^{92}_{39}\mathrm{Y} \longrightarrow {}^{92}_{40}\mathrm{Zr} + e^- + \bar{\nu}_e\ (+\gamma) \qquad \text{(Zr-92 is stable)} \tag{12}$$

9. **The betas (electrons) stop quickly and heat the fuel.**
   They are charged, so they stop within a short distance, like the fragments.

10. **Gammas and neutrons are neutral, so they travel centimeters before their energy turns into heat.**
    At the end, they pass their energy to a charged particle (an electron or a recoiling nucleus), which then heats the material locally.

11. **Antineutrinos escape, so only about $193\,\mathrm{MeV}$ per fission is recovered as heat.**
    That is $200\,\mathrm{MeV}$ minus the energy the neutrinos carry away. In SI units,

$$\gamma \approx 193\,\mathrm{MeV} \times 1.602 \times 10^{-13}\,\mathrm{J/MeV} \approx 3.1 \times 10^{-11}\,\mathrm{J/fission}. \tag{13}$$

12. **The 2–3 neutrons slow down, some cause new fissions, and this repeating cycle is the chain reaction.**
    This keeps the reactor running, and it is the topic of the next section, *Neutron Multiplication*.

Summary

Table 1 collects where the $\sim 200\,\mathrm{MeV}$ of one fission goes. Charged carriers (fragments and betas) deposit their energy right where the fission happens; neutral carriers (neutrons and gammas) spread it over centimeters; antineutrinos leave, which is why about $193\,\mathrm{MeV}$ is recovered as heat.

Carrier	Approx. energy (MeV)	Recovered as heat?
Fission fragment kinetic energy	~ 168	Yes, at the fission site
Prompt neutrons	~ 5	Yes, within cm
Prompt gammas	~ 7	Yes, within cm
Delayed betas	~ 8	Yes, locally
Delayed gammas	~ 7	Yes, within cm
Antineutrinos	~ 12	No, they escape

Table 1: Approximate distribution of the energy released in U-235 fission (typical textbook values).